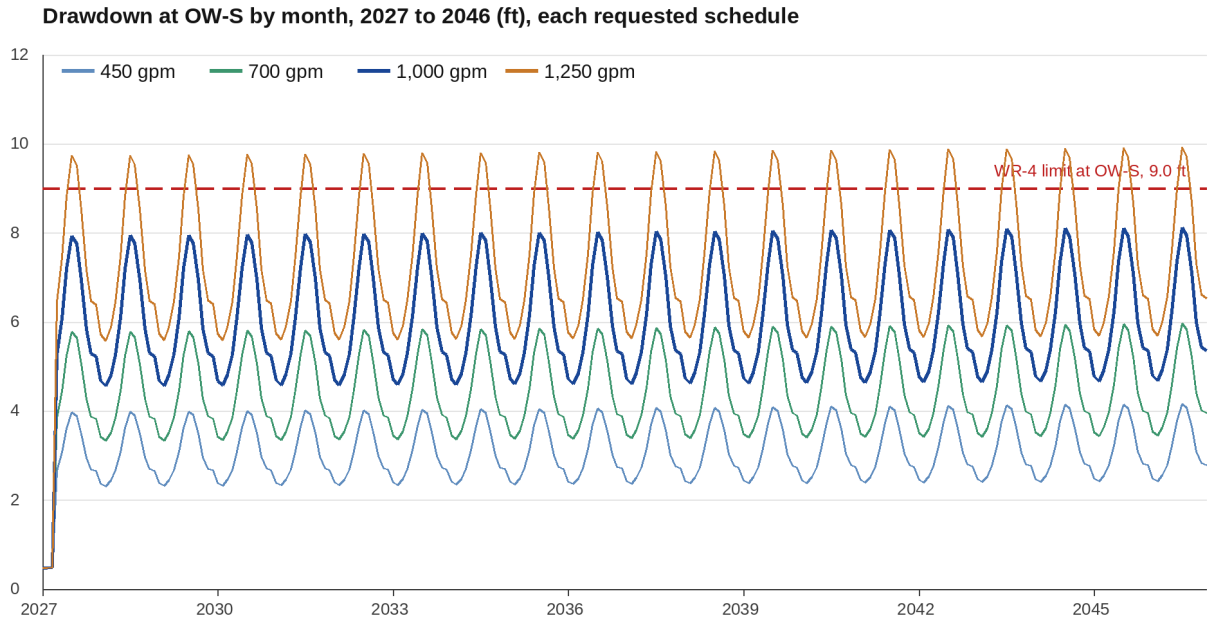
Reidel Road Well Field: Rule WR-4 Demonstration at Sentinel Well OW-S

Harlow Springs Utilities, Water Division. For the Corrin Valley Water Conservancy District, with the Reidel Road application. 6 November 2026.

The district will permit the 1,000 gpm schedule. Its greatest monthly drawdown at OW-S in 2046 is 8 ft against the Table 1 limit of 9.0 ft; the 1,250 gpm schedule reaches 10 ft.

Schedule (annual average)	Greatest monthly drawdown at OW-S, 2046	Within 9.0 ft
450 gpm	4 ft	yes
700 gpm	6 ft	yes
1,000 gpm	8 ft	yes
1,250 gpm	10 ft	no



Each line is the drawdown at OW-S in every month from January 2027 to December 2046 from the Reidel Road schedule and the Mill Street wells at their filed projection. Every schedule's greatest month falls in 2046.

At DW-17, the registered domestic well nearest the field (Rule WR-4 4.4), the 1,000 gpm schedule's greatest monthly drawdown in 2046 is 7 ft.

The aquifer test

The 72-hour test of PW-1 (14 to 17 September 2026, 1005 gpm on the totalizer) fits the Theis solution at OW-1, OW-2 and OW-3 to 0.08 ft: transmissivity 8,160 ft2/day, storativity 1.0e-03. Over three days the test cannot see what limits the drawdown over twenty years: carried forward unchanged, the same fit puts the Mill Street wells' 2025 drawdown at their monitoring well near 12 ft.

The Mill Street record

The monitoring readings since 2005 rise and fall with the region's wet and dry years, and the 2010s and early 2020s were wet against the 2005 to 2008 baseline. CV-07, the nearest network well in the aquifer, sits inside the Mill Street cone: its departure from the far network wells deepens with each step in Mill Street pumping and follows the summer peak. CV-02 and CV-11, beyond the cone, move together and moved with OW-S before pumping began, so they carry the natural level. Against it the Mill Street wells drew MS-MW down 5 ft on average in 2025.

The far field

With the valley walls at the mapped edges of the saturated sand and gravel on Plate 3 (5,000 ft apart) and the test's transmissivity and storativity, leakage through the till with a leakage factor of 3,526 ft reproduces the drawdown at MS-MW, CV-07 and OW-S from March 2009 to September 2026. Read against CV-07 or the wells' own 2005 to 2008 levels, the same record would ask for a leakage factor near 1,900 ft and put every schedule under 4 ft at OW-S, crediting the wet years to leakage.

Schedules

Reidel Road begins in April 2027, shared equally by PW-1 and PW-2, at each schedule's monthly factors; the Mill Street wells follow the projection filed in the 2026 water supply plan update (56 per cent TW-1, 44 per cent TW-2). The monthly working is in reidel_road_sentinel_forecast.xlsx.